

Deepak Mallampati

MACHINE LEARNING / MLOPS ENGINEER

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US-based, F-1 OPT (open to relocation/sponsorship)

PROFESSIONAL SUMMARY

ML engineer with 4+ years owning the end-to-end machine learning lifecycle - from data pipelines and training to deployment, real-time inference, and production monitoring. Builds robust, scalable ML and NLP systems with strong MLOps discipline (MLflow, Weights & Biases, CI/CD for ML, observability), and has cut p95 inference latency from 1.5s to 800ms through prompt compression, semantic caching, and model routing. Founder mindset: ships and maintains products solo, goes deep on AI infrastructure, and thrives in fast startup iteration.

CORE COMPETENCIES

ML Lifecycle Ownership • MLOps (MLflow, W&B, CI/CD for ML) • Real-Time Inference & Latency • NLP & Deep Learning • Data Pipelines & Orchestration • Python (Go-ramping) • Monitoring & Observability • Speech & LLM Systems

ML & NLP: PyTorch, TensorFlow, Keras | BERT/RoBERTa, NER, classification, summarization | Transformers | Fine-tuning (LoRA, QLoRA, PEFT) | Drift detection

MLOps & Infra: MLflow, Weights & Biases | Docker, Kubernetes | CI/CD (Jenkins, Azure DevOps) | Model monitoring & observability | AWS (SageMaker, Bedrock), Azure, GCP Vertex

Inference & Retrieval: Prompt compression, semantic caching, model routing | FAISS, Pinecone, Weaviate | Cross-encoder re-ranking | RAGAS, BERTScore eval

Data Engineering: Spark, Databricks, Kafka | PostgreSQL, MongoDB, Redis | Pandas, NumPy | Data lineage & governance

Languages: Python, SQL, JavaScript/TypeScript, Java (Go: ramping) | FastAPI

PROFESSIONAL EXPERIENCE

AI Engineer - Progress Solutions

Jul 2025 - Present | USA

- Owned the full lifecycle of production **agentic LLM systems** on a conductor-subagent framework - reducing token consumption by **42%** while sustaining accuracy across multi-step workflows.
- Established a structured **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to benchmark iterations and surface regressions pre-deployment.
- Optimized inference cost and latency via **prompt compression, semantic caching, and model routing**, sustaining SLA targets while reducing per-query overhead.
- Built fault-tolerant **data/automation pipelines** with scheduled orchestration and exponential-backoff retry, reducing failed production runs by **60%**.

AI Engineer Intern - Vanguard

Jan 2024 - Jan 2025 | USA

- Built **12 APIs and microservices** for real-time AI serving, reducing p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ teams.
- Evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success by **27%** and informing model-selection; engineered safeguards reducing unsafe outputs by **42%**.
- Developed AI agents and backend services across **25+ agents** integrated with production data pipelines and observability tooling.

ML & Gen AI Research Assistant - Kent State University

Oct 2023 - Jan 2024 | USA

- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, differential privacy), cutting re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned **Qwen3-72B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation; authored an IEEE-format paper.

ML Trainee - Emerson

Jun 2022 - Apr 2023 | India

- Fine-tuned **BERT and RoBERTa** for entity extraction and classification on domain data, achieving **89% F1** on custom NER.
- Built regression/classification models for equipment-failure prediction, anomaly detection, and forecasting with full data preprocessing pipelines.

ML Intern - Suvidha Foundation

Jan 2022 - Mar 2022 | India

- Built an end-to-end **speech-to-text + transformer summarization** pipeline across **5,000+ videos**, reducing manual editing effort by **70%**.

PROJECTS

Story Director LLM

Fine-tuned language model generating structured short-form video narratives, paired with an automated rendering pipeline - full model-to-production ownership.

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across **29+ sites** - owned end-to-end, real users.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0